Design, Development, Acceptance Testing, and Reiteration of the

GRAVIDforAMEN Maternal Health App in the context of Limpopo

Computer Science Honours Project Literature Review

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ABSTRACT

Maternal healthcare in rural contexts like Limpopo, South Africa, is currently characterised by a critical "socio-technical gap" where clinical information is often available. This information is described however as "tone-deaf" to local socio-economic realities [3, 5]. This review identifies significant knowledge gaps within these communities, particularly regarding maternal mental health and the healthy use of available indigenous foods. These are priorities that are frequently overshadowed by expensive, westernised dietary recommendations [4, 15]. While digital interventions offer a potential bridge, success is dampened by low digital literacy, with 66.2% of community members unable to use the internet independently, and structural barriers like high data costs and poor network reception [7, 12].

An evaluation of existing software reveals a divide between universal, low-cost accessibility (MomConnect) and high-specification, data-heavy functionality (Pregnancy+). Neither of these applications fully satisfy the community's need for "connectedness" and interactive support [11, 13]. For the GRAVIDforAMEN project, these findings necessitate a shift toward user-centred and participatory design frameworks that prioritize pictorial, voice-guided interfaces and "offline-first" functionality [16]. To ensure successful User Acceptance Testing (UAT), the project must explicitly foster "co-design readiness". This will be completed by utilizing trusted intermediaries and cultural protocols to mitigate power imbalances and ensure the resulting software is contextually situated and user-group centred [14,16].

CCS CONCEPTS

Maternal and Child health care • Digital Health • Sub-Saharan Africa (SSA), Limpopo • User Acceptance testing • Full stack development

KEYWORDS

Maternal health, digital interventions, Limpopo, mobile health apps, (UAT) user acceptance testing, Full stack development

1 Introduction

Maternal nutrition remains a critical challenge in Sub-Saharan Africa. This project focuses specifically on rural contexts such as Limpopo, where cultural beliefs, limited healthcare infrastructure, and food insecurity converge to shape pregnancy health and outcomes. The GRAVIDforAMEN mobile health application has been proposed as a culturally sensitive, offline-first tool designed to empower pregnant women and their families with accessible nutrition knowledge. It will do this while promoting maternal autonomy in nutrition decisions. In addition, it seeks to extend the supportive functions traditionally provided by Community Health Workers and to provide guidance on home gardening initiatives through partnerships with GIY (Grow It Yourself) South Africa. The scope of this project encompasses the design and development of the application, and the iterative adaptation informed by thematic analysis of Focus Group Discussions (FGDs) and workshops conducted in Limpopo. Accordingly, this literature review will examine maternal health contexts, the Socio-Technical landscape, design principles and features of mobile health interventions currently in practice, frameworks for user acceptance testing, and participatory workshops and engagement with key user groups. By structuring the review around these themes, the discussion establishes both the academic foundation and practical justification for the GRAVIDforAMEN app's lifecycle, ensuring that each stage of the project is evidence-based and aligned with local realities.

2 Body

2.1 Maternal Health Context in Limpopo

2.1 Overview and Background Context of the Environment

Sub-Saharan Africa's Maternal Mortality Ratio (MMR) sits at more than double the global average at 536 deaths per 100,000 live births [10]. Maternal healthcare in Limpopo is marked by systemic strain that shapes the attitudes of both providers and recipients.

Midwives and Community Health Workers (CHWs) work in frustration and exhaustion, facing severe shortages of resources ranging from birthing beds to soap and surgical gloves [6,8]. Scarcity forces staff to rely on manual monitoring in overcrowded units, delaying detection of life-threatening conditions [6]. A lack of infrastructure, training programs, and extremely short-staffed environments makes it impossible to perform maternal observations as they were traditionally taught [8,15]. As a result, patient interactions are often characterized by harsh, non-empathetic language, undermining trust in clinical care. Although staff explicitly advocate for continuous in-service education to master protocols and avoid mistakes, these requests are not being met [8].

Mothers' mindsets are shaped by both clinical advice and traditional belief systems. Many value medical guidance but are discouraged by long queues and potentially negative staff interactions (leading some to view antenatal care (ANC) as non-essential) [11,9]. Postpartum women often report feeling overwhelmed, depressed, and anxious, but maternal mental health remains severely underrepresented in research and digital interventions. It accounts for only 3.5% of focus in low- and middle-income countries [15]. In this context of scarcity, mothers may prioritize immediate welfare and income over digital tools, questioning the necessity of mobile applications [5].

Cultural myths or misconceptions may reinforce these attitudes. Fear of "bewitchment" by family enemies discourages early pregnancy disclosure, delaying clinic registration beyond 20 weeks [9]. Traditional healers are often preferred for initial symptoms such as morning sickness or foetal rotation [11]. Some networks encourage newborns to be given "holy water" before breastfeeding which can lead to complications [4]. Cultural protocols like *hlonipha* (respect/politeness) further limit women's willingness to initiate sensitive health discussions without guidance from elders or male relatives [14]. Finally, interventions often fail by assuming nuclear family structures and excluding grandmothers and fathers who remain structural elements in maternal care and decision-making [15].

2.2 The Socio-Technical Landscape: Infrastructure, Access, and Literacy in Limpopo

2.2.1 Constraints and Status of Technical Infrastructure

In rural Limpopo, mobile phones are the main gateway to health information, as radio and TV reception are often unavailable and computer ownership remains negligible at around 3% [4], but this access is even undermined by weak broadband infrastructure. Mwansa et al. [2025] report that 52.3% of rural residents lack consistent internet due to the absence of fibre-optic services and public Wi-Fi, leaving them reliant on basic 3G networks which are unreliable connections, with weak signals and geographic isolation across providers [11,12]. Internet reliability is further compromised by unstable electricity as frequent load-shedding disrupts towers and prevents device charging for days at a time [12,16].

Mothers' ability to use digital applications is largely influenced by financial resources. With mobile data costs being high, many only use the internet briefly after purchasing data at the end of the month [5]. This creates "digital exclusion," where data must be rationed between health apps and other needs. Muthelo et al. [2023] notes that mothers often "run short of data," undermining continuous monitoring via high-specification apps like Pregnancy+ [11].

Device distribution is also very varied and unequal. Seebregts et al. [2018] states that while

there is access across South Africa to mobile devices, socioeconomically disadvantaged communities may only have limited access to basic entry level devices that cannot support data heavy applications with any reliability [12,13]. Mothers also struggle with storage limitations, often deleting personal photos and other media to install health apps or failing to download them in the first place [4]. Communities will continue to prioritize familiar, accessible technologies such as radio and SMS over complex mobile applications that their infrastructure and hardware cannot steadfastly support [3,11].

2.2.2 Knowledge Access as a Human Rights Violation and Addressing the Knowledge Gaps.

In rural contexts, poor radio and television reception cuts off communities from critical health updates, which Maphosa [2021] frames as a "gross human rights abuse" and violation of fundamental freedoms [7]. This information poverty is compounded by knowledge gaps, mostly around maternal mental health and identified as a neglected priority in both literature and digital interventions. Mothers report feeling "overwhelmed" and "anxious" but lack any tools beyond clinical facts to manage these stressors [3]. Local studies highlight maternal mental health as underrepresented, Ameyaw et al. [2024] confirms that globally, mHealth is very effective for managing maternal anxiety and depression. This contrast suggests the gap in Limpopo stems not from technological interventions being ineffective in providing aid but from failing to adapt proven global interventions into culturally and linguistically appropriate local strategies [1]. There is a gap for future developments, and it is important that it is explored.

Klingberg et al. and Coleman et al. note that existing health information is often "too text-heavy," "generalized," and "tone-deaf" to local realities. Nutrition knowledge is a critical gap: women often lack the guidance on how to use indigenous foods, that are more easily accessible, during pregnancy, instead being presented with "ideal" diets featuring costly meats or fruits. This then leads to avoidance of nutrient-rich staples for example millet due to conflicting beliefs or limited nutritional knowledge [4]. To bridge the gaps, mothers emphasize they do not want a "digital textbook" but are searching for "connectedness" from interactive tools such as WhatsApp. This will enable peer support, community leaders, and "trusted voices" to foster shared understanding and reciprocity [3].

2.2.3 Digital Literacy Levels and Barriers to Work Around

Digital literacy in rural Limpopo is a major barrier as about 66% of community members are lacking the skills to use the internet independently [4,12]. This low competency means mobile phone use is restricted to basic functions, leaving a lot of mothers unable to navigate complex health applications without constant assistance [4]. To accommodate these levels, there is a recommendation to shift from text-heavy designs to pictorial, voice-guided interfaces that use local-language audio to reduce cognitive load [4].

Beyond design, literacy can be strengthened through community-centric interventions such as "digital bootcamps" or identifying "digital champions" within health units that can provide guidance to others, which can help build long term user confidence [2,11].

2.3 Theoretical Frameworks for System Design and Technology Acceptance

2.3.1 The Importance of User-Centred Design (UCD)

User-Centred Design (UCD) helps in mitigating the power imbalances between the researcher and user groups influence over the final application by including the voices of communities with first-hand knowledge of local challenges [16]. This approach works to shift digital health interventions from top-down, expert-led functionality to a "parent-centred" agenda aligned with the lived realities of beneficiaries and in turn creating a successful and specific to the user groups application [3]. To move beyond the static instruction, digital tools should employ modelling, demonstrating desired behaviours through others, which has proven effective for driving behaviour change in sub-

Saharan Africa [10]. Effective UCD methods that act partly as modelling tools include Challenge and Design Cards, which graphically simplify complex concepts for users with low levels of literacy [3], and technology probes that allow participants to effectively experience systems before giving feedback [16]. Unconstrained "blue sky" ideation can overwhelm community members unfamiliar with technology making it far less effective [16].

With regards to data collection two main methods seemed prevalent. Focus groups, which are useful for exploring cultural, food, and household dynamics [4], and semi-structured interviews which expose rich, flexible accounts of specifically discussed topics [5]. The most effective method is combining both interviews and focus groups is recommended to capture broader and more detailed insights simultaneously into digital health needs [11].

The research scoping group should extend beyond mothers to include fathers, who mostly make household food purchasing decisions [4], and grandmothers/elders, who remain a central player in maternal care and can override clinical advice with traditional beliefs [15]. Community health workers (CHWs) and community leaders must also be included because they act as the trusted intermediaries or "gatekeepers" and will add deeper levels of comfort to the environment [2,11]. Stakeholders in maternal healthcare are a diverse group of people and where possible researchers must encompass not only families but also government officials and partners to ensure supportive behaviour change at all levels [10]

2.3.2 Strategies for Engagement: Participatory Design and Trusted Intermediaries

Successful co-design in rural South Africa heavily depends on co-design readiness. This structure must be planned across emotional, cultural, methodological, and technological dimensions for each project uniquely [16].

Emotional readiness is built up through establishing reciprocal empathy, by moving from superficial observation to a deeper engagement tactic [16].

Cultural readiness requires an understanding and sensitivity to "invisible" norms, such as the experienced example of hesitation to cut fabric linked to traditional healers from the source [16].

Methodological readiness involves adapting workshop rhythms; rural settings often need flexible and extended timeframes to effectively accommodate cultural protocols and relationship building [14,16]. Understanding how the community hierarchies can alter group dynamics is a valuable tool to pick up on and use especially in situations where participants tend to defer to senior leaders [14,16]. Cultural phenomena like *hlonipha* (politeness protocols) impose gender-based speech restrictions and sometimes require women to obtain permission from the male participants before discussing sensitive topics such as postpartum care [14]. Trusted intermediaries that understand the cultural environment are therefore vital to create culturally safe spaces and encourage engagement from the more marginalised voices [14,16].

Technological readiness addresses the "blank page problem", where a lack of boundaries or instructions can overwhelm communities that are unfamiliar with the technology [16]. Instead of "blue sky" ideation, technology probes should be used to demystify systems through showing internal components or source code [16]. Specific, repetitive training builds confidence and familiarity, enabling participants to act as equal design partners in the process [16]. Tools such as Challenge and Ideation Design Cards, mentioned in 2.3.1, reduce the cognitive load and allow low-literacy users to tell stories without the pressure of drawing [3,14,16]. However, these cards can be so intuitive that participants sometimes skip brainstorming and move directly to prototyping the systems, risking shallow outputs if prototypes just replicate the cards [14,16]. Trust is the main goal to work towards because once you achieve a

level of trust you gain real insight into the community needs. Trust is demonstrated when participants shift from trying to please researchers to advocating for their own needs [16].

2.4 Design and Implementation Paradigms for Rural Digital Interventions

2.4.1 User Interface (UI) and Experience (UX): Designing for Low-Literacy and High-Constraint Environments

Designing applications for high-constraint environments, like Limpopo, requires moving away from text-heavy "westernised" user interfaces and toward pictorial, voice-guided, and multilingual frameworks [3,16]. The standard text-based communication is also often perceived as "too generalised" or challenging to navigate, leading researchers to prioritise voice messaging in the local languages such as Sepedi [4,11]. Voice delivery improves comprehension levels among users with limited formal education. They also help remove literacy barriers and allow mothers to listen to instructions while performing household chores [4,11].

To reduce cognitive load, successful designs tend to use realistic pictorials and large, intuitive icons that represent recognisable items from the users' environments and family structures [3,14]. These visuals act to foster shared understanding without reliance on technical jargon [16].

Two critical metrics that should be used for rural interventions include learnability and memorability as they are essential for users with intermittent access due to data costs or power outages [7]. Learnability refers to how easily a system can be understood during the first use, while memorability is the ability to recall functions after periods of non-use [7]. High-performing applications make sure to avoid steep learning curves by employing consistent navigation and simplified menus setups [7]. Finally, to support users with low digital literacy, systems may incorporate digital champions within clinics to guide new users and support confidence use of the technology [2,4].

2.4.2 Comparative Utility and Feature Requirements: Evaluating MomConnect vs. Pregnancy+ for Rural Adoption

Drawing on the Technology Acceptance Model (TAM), Perceived Usefulness (PU) is the main driver of technology adoption in rural contexts; when mothers see that there is tangible utility, they develop the "motivational access" needed to overcome the structural barriers otherwise limiting them [12]. In South Africa, MomConnect and Pregnancy+ represent two contrasting maternal health software-utility styles.

MomConnect is valued for universal accessibility and uses SMS and USSD menus with no data costs, making it reliable for clinic registration and foetal development reminders [13]. However, Muthelo et al. [2023] notes its one-way messaging feels limited, cold and "text-heavy," failing to provide the "connectedness" or chatroom environment that the mothers mother's desire [3,11].

Pregnancy+ offers higher-specification features including 3D models and daily tracking that boosts the first-time mothers' confidence but requires a lot of data and large storage capacity [11]. Ameyaw et al. [2024] highlights that advanced systems are often unusable for rural women due to the low literacy and shared device practices, which compromise privacy. Muthelo et al. [2023] further found mothers were forced to delete personal photos and media to make space for the software or run out of data very quickly, excluding the most socio-economically vulnerable [11].

The review suggests mHealth in low- and middle-income areas must move away from the higher-income model of individualised smart phone use and instead focus on locally based software in varied languages that also reflect the family decision-making structures [1]. Current interventions are often "tone-deaf" to local realities and designed for nuclear families and exclude influential household decision-makers from the software [3,16]. To drive adoption, future systems should incorporate Family Modes that include fathers or other family members,

especially the primary food purchasers [4], and grandmothers, who can override digital advice with traditional beliefs [14].

Applications must also provide some form of tangible benefits beyond clinical facts. This includes myth-busting features to address context relevant taboos such as fears of “bewitchment” or reliance on traditional healers [11]. Downs et al. [2019] and Klingberg et al. [2022] suggest that interventions in the future should link health information with essential services like home gardening or agricultural production to improve nutrition, and group saving schemes (stokvels) and in turn ease financial stressors [4,5]. This concept will be implemented in the GRAVIDforAMEN application. At the end of the day, rural mothers seek not a textbook or all-knowing fancy application but merely a multilingual, voice-guided platform that fosters reciprocity and shared understanding within community structures [3,11].

3 Conclusions

This literature review highlights that the effectiveness of digital maternal health interventions in rural South Africa does not only depend on the accessibility of data but on the "tangible utility" and the cultural resonance of the system itself [2, 12]. The research stresses three primary takeaways:

Priority Realignment: Mothers prioritize mental health support and a more localised nutritional advice system over static clinical facts, they want "chatrooms" and peer connection rather than digital textbooks [3].

Design for Constraint: The high levels of digital illiteracy, infrastructure failures, and financial instability require a transition from text-heavy westernised interfaces to pictorial, multilingual, and voice-guided frameworks that reduce the cognitive load of the user and allow for effective and easy use of the software [4, 11].

Co-design Readiness: Successful implementation is predicated on the emotional and cultural readiness of the project. This involves long-term building of trust, the inclusion of "Family Modes" for fathers and elders, the inclusion of the trusted intermediaries, and the use of technology probes and design artefacts to demystify complex systems before their deployment [2, 16].

The path toward User Acceptance Testing (UAT) and deployment for the GRAVIDforAMEN application must move beyond generic evaluation to active advocacy by the community [16]. By implementing an offline-first, culturally sensitive architecture that also accommodates the specific temporal rhythms and communication protocols of Limpopo, the project can provide a sustainable platform that provides guidance and care for the mothers that need it most.

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